



In Silico Identification and Characterization of Rare Cold Inducible 2 (RCI2) Gene Family in Cotton

Muhammad Fahad¹ · Leeza Tariq² · Muhammad Tanveer Altaf³ ·
Muhammad Shahnawaz⁴ · Mudassir Aslam⁴ · Waqas Liaqat⁵ · Inayat Ullah⁶ ·
Izhar Ullah⁷ · Heba I. Mohamed⁸ · Abdul Basit⁹

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Abstract

RCI2/PMP3s are involved in biotic and abiotic stresses and have an influence on the regulation of many genes. RCI2/PMP3 genes, which particularly encode small membrane proteins of the PMP3 family, are involved in abiotic stress responses in plants. In this work, in silico studies were used to investigate RCI2's potential function in stress tolerance and organogenesis. We conducted an extensive study of the RCI2 gene family and revealed 36 RCI2 genes from cotton species that were distributed over 36 chromosomes of the cotton genome. Functional and phylogenetic examination of the RCI2/PMP3 gene family has been studied in Arabidopsis, but in cotton, the RCI2/PMP3 genes have not yet been studied. Phylogenetic and sequencing studies revealed that cotton RCI2s are conserved, with most of them categorized into six distinct clades. A chromosome distribution and localization study indicated that cotton RCI2 genes were distributed unevenly on 36 chromosomes with segmental duplications, suggesting that the cotton RCI2 family is evolutionarily conserved. Many *cis*-elements related to stress responsiveness, development, and hormone responsiveness were detected in the promoter regions of the cotton RCI2. Moreover, the 36 cotton RCI2s revealed tissue-specific expression patterns in the development of cotton performed by transcriptome analysis. Gene structure analysis indicated that nearly all RCI2 genes have two exons and one intron. All of the cotton RCI2 genes were highly sensitive to drought, abscisic acid, salt, and cold treatments, demonstrating that they may be employed as genetic objects to produce stress-resistant plants.

Keywords Cotton RCI2 · PMP3 · Phylogenetic analysis · Genome-wide · Tissue expression pattern · Stress responses

Abbreviations

RCI2 Rare cold inducible 2
PMP3 Plasma membrane protein 3

Lti	Low temperature induced
MW	Molecular weight
CDS	Coding sequence
Gdna	Genomic DNA

Introduction

Plants are severely affected by both biotic and abiotic conditions such as salinity, drought, low temperatures, pests, and diseases, among others to survive in these conditions, plants change their external and internal processes, including their physiological, molecular, and biochemical pathways, for successful adaptation in stressful and harsh environments (Ashry et al. 2018; El-Mahdy et al. 2021). The cotton crop is also affected by abiotic stresses such as salinity, extreme temperature, drought, and soil pH (Mohamed and Abdel-Hamid 2013). Additionally, research on the activities of multiple genes and their functions in improving stress pressures, particularly under high temperatures and salinity, may provide further information on cotton under stressful conditions (Ullah et al. 2022). Abiotic stresses have an immediate impact on the cell wall composition and plasma membrane, which serve as a barrier between the cell and the outer environment (Sachdev et al. 2023; Mohamed et al. 2018; Panjabi-Sabharwal et al. 2010).

The plasma membrane is critical for maintaining cell integrity, cellular osmoses, and signal transduction. Abiotic stress exposure increases the permeability of the plasma membrane, thus permitting the loss of electrolytes from the cell (Lyons 2012; Abu-Shahba et al. 2022; Khan et al. 2023). As a result, plasma membrane safety is critical when exposed to abiotic stress. In cell–cell interaction, ion transport, and signal transduction, certain integral membrane proteins play critical roles (Marmagne et al. 2004; Banik and Dutta 2023). Numerous vital membrane proteins play a crucial role in facilitating the transport of ions and other molecules across the membrane; simultaneously, they contribute to maintaining membrane potential and restoring ion homeostasis in response to diverse environmental stresses. Moreover, transcription factors regulate downstream gene expression by binding to specific *cis*-acting regions within the promoter, where they exert *trans*-activating or *trans*-repressing effects (Fahad et al. 2023a). Among these proteins, Rare Cold Inducible 2 (RCI2) proteins were involved in ion homeostasis and performed functions for the maintenance of plasma membrane potential (Navarre and Goffeau 2000; da Silva et al. 2023).

Plasma membrane proteins 3 (PMP3), first discovered in *Saccharomyces cerevisiae*, are a gene engaged in reducing membrane stability due to low temperatures and salt intensity (Medina et al. 2001). PMP3 homologs known as rare cold-inducible (RCI) and low temperature (LTI) genes have been discovered to exhibit their expression under abiotic conditions in many plant species, e.g., *Arabidopsis thaliana* (Khurana et al. 2015). Proteins of the PMP3 family are named after the 55aa residue long plasma membrane protein of yeast and are produced by the PMP3/Sna1 gene (Navarre and Goffeau 2000). Plant RCI2/PMP3 are very similar in size, having 50 to 80 aa residues, structure, and sequence of PMP3 as

well as containing two extremely hydrophobic domains separated through a short sequence loop and being an integral part of membrane proteins. RCI2/PMP3s are believed to play a role in cell ion homeostasis by an unclear mechanism, as shown by the yeast PMP3 proteins (Navarre and Goffeau 2000). RCI2 genes mostly contain an intron and two exons that encode low-molecular-weight (MW) and most of the hydrophobic proteins through the domain (PF01679) of PMP3 (Rocha 2016; Sun et al. 2022).

RCI2 proteins of plants are grouped into two classes depending on the C-terminal tails and have two putative α -helix domains. RCI2 in group I are short proteins (51–64 aa) while in group II are longer proteins (59–89 aa) as compared to group I by hydrophilic C-terminal extended and therefore are enriched in charged residue (Medina et al. 2007). Plant species, fungi, animals, and prokaryotes also have RCI2 family genes (Medina et al. 2007; Zhao et al. 2014a, b). Many plants possess RCI2 family genes such as the low-temperature-inducible genes OsLti6A and OsLti6B in *Oryza sativa* (Morsy et al. 2005) wpi6 in *Triticum aestivum* (Imai et al. 2005), BLT101.1 and BLT101.2 in barley (Goddard et al. 1993), ESI3 in *Lophopyrum elongatum* (Galvez et al. 1993), ZmPMP3s and ZmPMP3-6 in *Zea mays* (Fu et al. 2012), PpSHP1 and PpSHP2 in *Physcomitrella patens* (Kroemer et al. 2004), PutPMP3-I and PutPMP3-II in *Puccinellia tenuiflora* (Zhang et al. 2008), CsRCI2A in *Camelina sativa* (Kim et al. 2016), MtRCI2A, MtRCI2B and MtRCI2C in *Medicago truncatula* (Long et al. 2015), MsRCI2A from *Medicago sativa* (Long et al. 2015) indicating that all homolog have related roles in controlling ions homeostasis and regulating membrane potential in the presence of salt stress conditions.

It has been proposed that AtRCI2A and AtRCI2B act as stabilizers of membrane proteins under water stress caused by chilling or other external factors that reduce water supply (Navarre and Goffeau 2000; Zhao et al. 2014a, b). The expression of RCI2 genes has been associated with Na^+ ion uptake in cells (Navarre and Goffeau 2000; Zhang et al. 2022). It was previously reported that the PMP3 deletion mutant in yeast accumulates extra Na^+ ions and is more susceptible to salt stress (Navarre and Goffeau 2000; Zhao et al. 2014a, b). PMP3-like genes are found in plants as gene families and are referred to as RCI2, PMP3, or Lti6 genes (da Silva et al. 2023). PMP3 genes have been identified and cloned from tissues exposed to the abiotic stresses (cold and salinity). Abiotic stress-mediated tissues (cold, salinity) were first used to isolate and clone PMP3-like genes (Kumari et al. 2017).

PMP3 genes are expressed in plants in response to a diverse range of abiotic conditions (Chidambaram et al. 2015; Brunetti et al. 2018). Although, these genes are generally present in the plasma membrane (Kumari et al. 2017), some plants show PMP3 localization in other subcellular organelles (Khurana et al. 2015; Medina et al. 2007). In this research, identification and characterization of cotton genes (GhRCI2) that show high resemblance with AtRCI2A and B are described. Furthermore, it is demonstrated that RCI2 related proteins are found in a diverse range of organisms, such as fungi, plants, animals, and prokaryotes, in which they are structured as gene families. The comparison between cotton RCI2 genes and AtRCI2 genes from Arabidopsis gave useful knowledge about the functional and organizational roles of RCI2 genes. Interestingly, such genes and their predicted proteins are extremely conserved at the structural and sequence levels, indicating a similar evolutionary history. The subcellular

localization of RCI2 proteins suggests that all of these are found at the plasma membrane (Kim et al. 2022).

Despite their similarities, conserved subcellular localization, and similar origin, functional and expressional analysis indicate that unique RCI2 proteins may play critical roles in response to biotic and abiotic stresses. According to these analyses, many aspects of RCI2 gene function and evolution in cotton are studied.

Materials and Methods

RCI2 Gene Family Identification and Sequence Analysis in Cotton

RCI2 gene family proteins of *A. thaliana*, *G. hirsutum*, *G. barbadense*, *G. raimondii*, and *G. arboreum* were downloaded from the Arabidopsis Information Resource (TAIR, <http://www.arabidopsis.org/>), and the Cotton Genomic Database (CottonFGD) (<https://cottonfgd.org/analyze/>), correspondingly, based on the protein IDs of Arabidopsis used in previous studies (Medina et al. 2007; Long et al. 2015). These sequences were then used as references in a BLASTP query of the genomic database (<https://cottonfgd.org/analyze/>) with an *e* value of $1e-10$. Additionally, the profile of the PMP3 domain (PF01679) was downloaded from the Pfam database (Protein Family: <https://pfam.xfam.org/>) and examined by domain analysis with SMART (<http://smart.embl-heidelberg.de/>) (Letunic et al. 2015). Finally, redundant sequences were also removed, resulting in 36 RCI2 protein sequences. Chromosomal location, exon number, and amino acid were also obtained from the cotton genome database. The ProtParam server (<http://web.expasy.org/protparam/>) was used to investigate the physicochemical properties of each RCI2 protein, such as molecular weight (MW), isoelectric point (pI), and grand average of hydropathy (GRAVY). Subcellular and cellular localization of the RCI2 gene family was predicted by an online program, TargetP 1.1 Server (<http://www.cbs.dtu.dk/services/TargetP-1.1/index.php>).

Phylogenetic Analysis

The full-length RCI2 protein sequences of *G. barbadense*, *G. raimondii*, *G. hirsutum*, and *G. arboreum* were used in the phylogenetic analysis which was aligned by ClustalW in the MEGA 6.0 program (Deokar et al. 2015). After that, phylogenetic analysis was performed using multiple sequence alignment of protein sequences and a tree was generated by using the Neighbor-Joining (NJ) approach with parameters such as Poisson Model, bootstrap value, *p*-distance, complete deletion, and Substitution. Moreover, a single phylogenetic tree was created using the RCI2 protein sequence of *G. hirsutum* for further analysis.

Chromosomal Mapping, Duplication Analysis, and Synteny Analysis

RCI2 genes position on chromosomes was measured by using genes features available in the CottonFGD database. Map Chart program was used for mapping RCI2

genes on chromosomes. BLASTP program was used to made for duplication study using protein sequences of *G. hirsutum* itself as aquery. Genes duplications were identified based on the similarity index greater than 80% and at least alignment percentage greater than 80% as compared with proteins total length (Yang et al. 2008). Clustal Omega software (<https://www.ebi.ac.uk/Tools/msa/clustalo/>) was used to calculate non-synonymous (Ka) and synonymous (Ks) substitution incidences for further assessment of duplication events (Sievers and Higgins 2018) and paired genes alignment was examined with PAL2NAL tool (<http://www.bork.embl.de/pal2nal/>) (Suyama et al. 2006). Moreover, syteny investigation of RCI2 protein sequences was used to make a circus plot to visualize the similarity pattern by using Circoletto (<http://tools.bat.infspire.org/circoletto/>) software (Darzentas 2010).

Domain and Conserved Motifs Analysis

To analyze domain structure, the polypeptide sequences were submitted to the Conserved Domain Database (CDD) platform of NCBI to predict conserved domain. The conserved domains were visualized by TBtools. MEME suite tool (version 4.8.2) (<http://www.nbc.net/>) was used to recognized conserved protein motifs in all gene IDs of cotton. The following parameters are used such as optimum width (6–450), the maximum number of motifs 25, and no. of reappearances (Bailey et al. 2009). The most common motif elements were also visualized by TBtools

Gene Structure and Subcellular Localization Analysis

RCI2 genomic and cDNA sequences were taken from online CottonFGD database. The online Gene Structure Display Server (GSDS 2.0) (<https://gsds.cbi.pku.edu.cn/>) was used to examine gene structure by matching each gene sequence through its associated cDNA sequence. Annotation and functional analysis of RCI2 genes were executed by with the CottonFGD database. RCI2 GO annotations were linked by GO terms by Biological Function, Molecular Function, and Cellular Component.

Promotor *cis*-Element Analysis

Cotton FGD database was used to extract nucleotide sequences of 2 kb from the start codon of 38 RCI2 genes for promotor examination. Then *cis*-regulatory elements of RCI2 genes were identified by using these sequences in plantcare software (<http://bioinformatics.psb.ugent.be/webtools/plantcare/html/>).

Expression Pattern of RCI2 Genes

RCI2 gene family expression patterns under different abiotic stresses such as cold, heat, drought and salt in several tissues were checked by RNA-seq data of *G. hirsutum* available in the CottonFGD database. Heatmap was generated by using FPKM values for RCI2 gene expression. The Gene IDs highly displayed in abiotic

conditions and different tissues were used to construct a Venn diagram using the online Genvenn software (<http://habtom.github.io/biojs-vis-genvenn/>).

RNA Extraction and qRT-PCR Analysis

Seeds of upland cotton (*Gossypium hirsutum* L.) were germinated in soil-based pots at 26/22 °C for 16/8 h. At 4 weeks post-germination, three-leaf stage cotton seedlings were subjected to abiotic stresses including cold (4 °C), salinity (350 mM NaCl solution) and ABA (200 mM) (Jian et al. 2023). Leaves were harvested at 3, 6 and 12 h post treatment for all stresses with three independent biological replicates per time point. Leaves were immediately frozen in liquid nitrogen and stored at – 80 °C prior to RNA extraction. Total RNA isolation was performed using RNA Prep Pure Plant Kit (TIANGEN Biotech, Beijing, China) and quantified using a NanoDrop 2000 spectrophotometer (Thermo Scientific, USA). Then, 1 μ l of total RNA was utilized for cDNA synthesis and qPCR was performed with 2 $\times$ SYBR Green PCR Master Mix using three biological replicates. The quantification results were analyzed by the $2^{-\Delta\Delta C_t}$ method using GhActin as the internal normalization control (Livak and Schmittgen 2001).

Results

Identification of RCI2 Genes in Cotton

For the identification of RCI2 genes in cotton, the protein sequence of PMP3 domain was attained from the Pfam database to search against the cotton genome database. Furthermore, Arabidopsis proteins sequence was used as queries to find the maximum cotton RCI2 proteins. After comprehensive analyses of the cotton genome database, 36 RCI2 protein sequences were discovered. The initial sites of the 36 prospective cotton RCI2s also were identified from the cotton genome database so each cotton RCI2 gene (RCI2-1 to RCI2-36) was labeled according to its pattern of distribution on the 36 chromosomes (RCI2-1 to RCI2-38). Physiochemical analysis of different parameters by using ExPASy indicated significant variations between cotton RCI2 proteins. The base pair length of *G. hirsutum* RCI2 genes ranged from 259 bp (Gh_A13G2310) to 439 bp (Gh_D07G1300), where protein sequence ranged from 57 to 78 aa with the pI values from 4.063 to 8.67, and the MW from 6.283 to 8.345 kDa (Table 1). Further, the values of GRAVY for *G. hirsutum* RCI2 proteins ranged from 0.415 (Gh_D07G1300) to 1.36 (Gh_A13G2310). The base pair length of barbadense RCI2 genes ranges from 403 bp (Gbar_D04G008790) to 2036 bp (Gbar_D13G020310), where protein ranging from 54 to 85 aa with the pI values from 3.778 to 7.971, and the MW from 5.935 to 9.919 kDa (Table 1). In addition, the GRAVY values of barbadense RCI2 proteins ranged from 0.582 (Gbar_D12G013000) to 1.633 (Gbar_D04G008750). The base pair length of arborium RCI2 genes ranges from 259 bp (Ga13G2237) to 439 bp (Ga07G1464), while protein ranging from 54 to 78 aa with the pI values from 3.778 to 8.851, and the MW

Table 1 Basic characteristics of RCI2 gene family in Cotton

ID position	Start, end	Length (aa)	PI	MW (kDa)	GRAVY	Subcellular pre- diction
Gh_A06G0886.1	33,798,337, 33,798,653	317	4.535	8.659	0.851	Plasma membrane
Gh_A07G1199.1	26,779,291 26,779,729	439	8.67	8.462	0.436	Endoplasmic reticulum
Gh_A08G1073.1	73,729,784 73,730,143	360	7.779	8.345	0.615	Plasma membrane
Gh_A13G1613.1	74,966,371 74,966,695	325	4.535	8.702	0.729	Plasma membrane
Gh_D05G0543.1	4,416,549 4,416,865	317	4.535	8.699	0.835	Plasma membrane
Gh_D06G1036.1	22,102,534 22,102,849	316	4.535	8.659	0.851	Plasma membrane
Gh_D07G1300.1	20,685,640 20,686,078	439	8.851	8.513	0.415	Chloroplast
Gh_D08G1354.1	44,407,305 44,407,664	360	7.779	8.345	0.615	Plasma membrane
Gh_D13G1917.1	54,509,119 54,509,377	259	4.063	6.269	1.354	Endoplasmic reticulum
Gh_D13G1973.1	55,083,763 55,084,076	314	4.535	8.668	0.742	Plasma membrane
Gbar_A05G005030	4,846,970 4,848,668	1699	4.535	8.699	0.835	Plasma membrane
Gbar_A06G010260	34,357,308 34,358,079	772	4.535	8.659	0.851	Plasma membrane
Gbar_A13G019570	103,250,714 103,252,223	1,510	4.063	6.283	1.36	Endoplasmic reticulum
Gbar_A13G020130	103,826,460 103,827,226	767	4.288	8.659	0.838	Plasma membrane
Gbar_D04G008750	17,040,222 17,041,211	990	3.778	5.935	1.633	Plasma membrane
Gbar_D04G008790	17,158,631 17,159,033	403	4.073	9.919	1.532	Endoplasmic reticulum
Gbar_D05G005480	4,443,021 4,444,556	1536	4.535	8.699	0.835	Plasma membrane
Gbar_D06G010770	21,921,879 21,922,672	794	4.535	8.659	0.851	Plasma membrane
Gbar_D12G013000	39,566,818 39,568,853	2036	7.971	8.034	0.582	Chloroplast
Gbar_D13G019720	54,098,281 54,099,891	1611	4.063	6.269	1.354	Endoplasmic reticulum
Gbar_D13G020310	54,640,242 54,640,994	753	4.535	8.668	0.742	Plasma membrane
Gorai.001G148000	20,275,276 20,275,980	705	8.851	8.495	0.449	Plasma membrane
Gorai.008G134100	38,109,938 38,112,301	2364	7.971	8.576	0.552	Chloroplast

Table 1 (continued)

ID position	Start, end	Length (aa)	PI	MW (kDa)	GRAVY	Subcellular prediction
Gorai.009G056100	4,080,648 4,081,415	768	4.535	8.713	0.836	Plasma Membrane
Gorai.010G113800	21,808,008 21,809,733	1726	5.565	8.681	0.855	Plasma membrane
Gorai.012G093000	16,478,253 16,478,790	538	3.778	5.935	1.633	Plasma membrane
Gorai.013G210800	52,271,613 52,273,507	1895	4.063	6.269	1.354	Endoplasmic reticulum
Gorai.013G215700	53,575,964 53,576,735	772	4.535	8.668	0.742	Plasma membrane
Ga05G0567	5,123,237 5,123,553	317	4.535	8.699	0.835	Plasma membrane
Ga06G1113	38,284,339 38,284,656	318	4.288	8.631	0.864	Plasma membrane
Ga07G1464	27,610,496 27,610,934	439	8.67	8.462	0.436	Endoplasmic reticulum
Ga13G2237	117,600,841 117,601,099	259	4.063	6.283	1.36	Endoplasmic Reticulum
Ga13G2292	118,182,480 118,182,797	318	4.535	8.702	0.729	Plasma membrane

from 5.935 to 8.702 kDa (Table 1). Moreover, the GRAVY values of arborium RC12 proteins ranged from 0.436 (Ga07G1464) to 1.633 (Ga04G1173). The base pair length of raimondii RC12 genes ranges from 538 bp (Gorai.012G093000) to 2364 bp (Gorai.008G134100), which protein ranging from 54 to 78 aa with the pI values from 3.778 to 8.67, and the MW from 5.935 to 8.713 kDa (Table 1). The GRAVY values of raimondii RC12 proteins ranged from 0.449 (Gorai.001G148000) to 1.633 (Gorai.012G093000), signifying that they are highly hydrophobic.

Phylogenetic and Gene Structure Analysis of RC12 Genes in Cotton

For better understanding of evolutionary relationship between cotton RC12 genes, a phylogenetic tree was constructed with the help of protein sequences of Arabidopsis and 36 cotton RC12 genes using MEGA 11 with the NJ method (Fig. 1). In this study 36 cotton RC12 proteins were classified into 4 clades such as clade A, clade B, clade C, and clade D. Clade A having 6 members present UPF0057 membrane protein involved in low temperature and salt responsive protein family and clade B having 17 members, clade C having 10 members, and clade D having three members, respectively (Fig. 1). The size of clades ranged between 3 to 17 genes in the clade D and clade B subgroups. *G. hirsutum* RC12 genes were dispersed between different subgroups such as clade A contains 2 genes, clade B contains 5 gene, clade C contains 5 genes. Phylogenetic tree analysis of subclasses was supported through

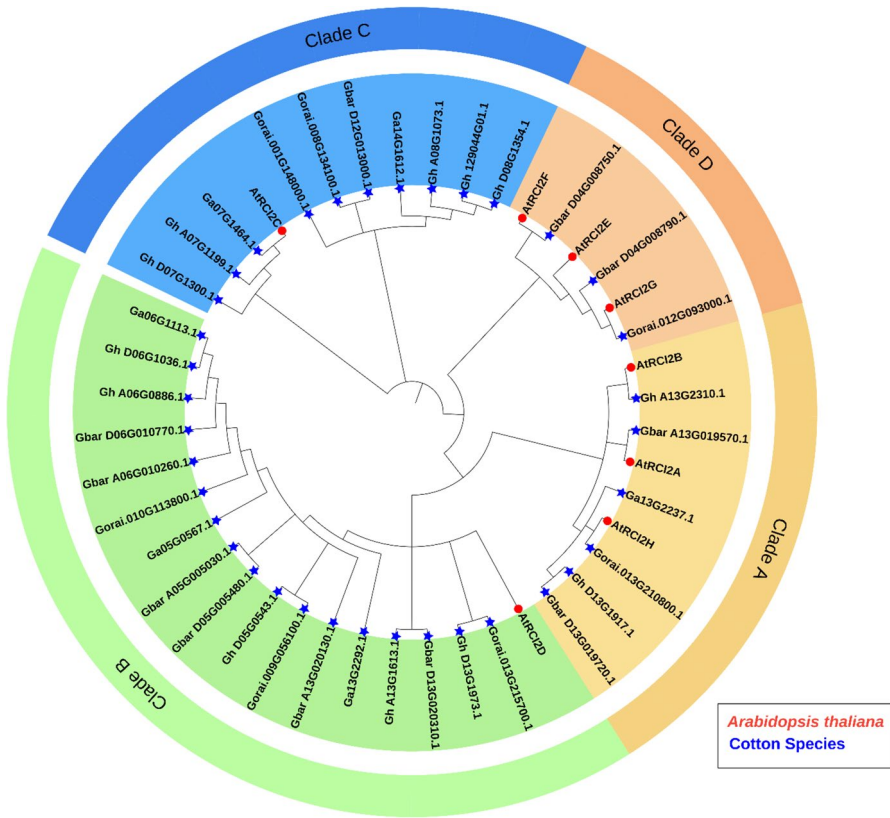


Fig. 1 Phylogenetic relationships among 44 RCI2 genes in both *Arabidopsis* and cotton species. The construction of the phylogenetic tree was executed through MEGA 11 software, employing the NJ method and utilizing 1000 bootstrap replicates. The resulting tree is categorized into four distinct clades, identified as clade A, clade B, clade C, and clade D

the relationship in the demonstration of RCI2 proteins domain repeats within the subclasses. Clade B subgroup in the phylogenetic tree was grouped at two different positions, clade B (I) having 7 genes while the clade B (II) cluster having 10 genes in the phylogenetic tree.

Chromosomal Mapping of RCI2 Genes and Syntenic Analysis

Cotton FGD database (<https://cottonfgd.org/>) provided the chromosomal rank of all 36 RCI2 members. As a result, 36 RCI2 genes were found on all upland cotton chromosomes (A04–A13 and D04–D13). On the other hand, one RCI2 gene (Gh_Sca129044G01.1) was not seem to be mapped to any chromosome but was identified on a scaffold. The genes were arranged unequally over chromosomes with varied abundances on particular chromosomes as well as chromosomal regions (Fig. 2). However, the arrangement of genes on chromosomes of *G. hirsutum* was similar

to that of D chromosomes expect D13 which has 2 genes. For *G. barbadense* the highest density of RCI2 genes was found on the A13 and its homolog D13 chromosome and also D04, with 2 genes per each chromosome, and the lowest density of RCI2 genes were reported on the remaining chromosomes, with just 1 gene. *G. arborium* and *raimondai* genome contains 1 gene on each chromosome expect chromosomes 13 in both have 2 genes. The D genome had a greater number of genes reported than the A genome. A duplication finding indicates that segmental duplication is the principal cause of RCI2 genes development in upland cotton. When duplicated genes are distributed on various chromosomes, this is known as segmental

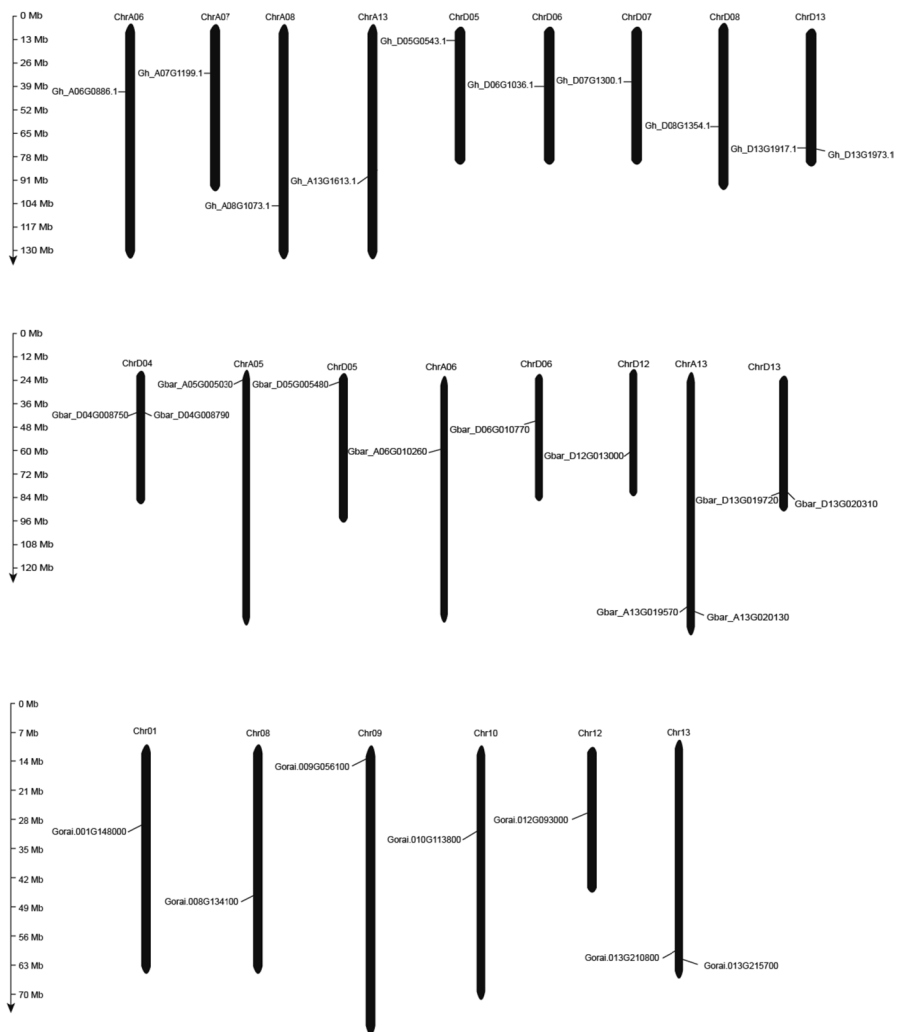


Fig. 2 Chromosomal locations of Cotton RCI2 genes. Gene positions and chromosome sizes are shown in Megabase (Mb). Distribution of genes in cotton chromosomes

duplication (Fig. 3). Duplication analysis revealed that segmental duplication is the leading cause of expansion of RCI2 genes family in cotton. Segmental duplication occurs when duplicated genes are located on different chromosomes. Circos plot was developed for synteny analysis using online software Circoletto in which arcs of different colors show varying percentage of similarity between the sequences (Fig. 3). Syntenic relationship between one tetraploid species *G. hirsutum*, *G. barbadense* and one diploid *G. hirsutum* was created through circos software. Different chromosomes of selected species are marked with different colors.

Domains Structures and Motif Distribution in RCI2 Genes

The structural mechanism and variation of genes and diversity of conserved motifs allow for the evolution of multigene families (Wang et al. 2013a, b). As a result, the evolutionary relationships of RCI2 genes were investigated using a method of motif distribution and domain structure analysis (Fig. 4). Conserved protein motifs were explored by MEME for 36 identified cotton RCI protein sequences. Four distinguished motifs were identified. Motif 2 is present in all cotton RCI genes. Motif 1 is conserved among all the RCI2 genes, and motif 2 is present in all genes except in 4 RCI2 genes. Similarly, motif 3 is conserved across all genes, except those in different species. Additionally, motif 4 is identified in 7 RCI2 genes. Notably, the domain analysis reveals the conserved domain of pmp3 in all RCI2 genes (Fig. 4).

Subcellular Localization and Gene Structure Analysis

The cotton RCI2 genes' biological activity, molecular function, and cellular components were investigated using the Gene Ontology (GO) database. The results indicated that 36 cotton RCI2 genes are involved in the cellular component and the part of the integral component of the plasma membrane. *G. hirsutum* RCI2 genes were greatly involved in the cellular component. In this research, RCI2 proteins of *G. hirsutum* eight genes were likely to be located in the plasma membrane, whereas three genes were predicted in endoplasmic reticulum (E.R) and one gene (Gh-D07G1300.1) was located in chloroplast thylakoid membrane. RCI2 of *G. barbadense* 7 genes were located in plasma membrane; three were located in endoplasmic reticulum and Gbar1213000.one gene in chloroplast thylakoid membrane (Table 1, Fig. 5). *G. arborium* RCI2 four genes were predicted to be located in plasma membrane; two were located in E.R. no one gene was predicted in chloroplast thylakoid membrane. Five RCI2 genes of *G. ramaindoi* were located in PM and one was in ER and also one gene was predicted in chloroplast thylakoid membrane. For more understanding into the evolution and structural diversity of the RCI2 family in each cotton species, we analyzed the gene structure by aligning the CDS sequences with the genomic DNA sequences of cotton RCI2 family. This study showed that the RCI2 domain's intron/exon composition is extremely conserved. According to their structural analysis, nearly all RCI2 cotton genes showed one intron and two exons (Fig. 6).

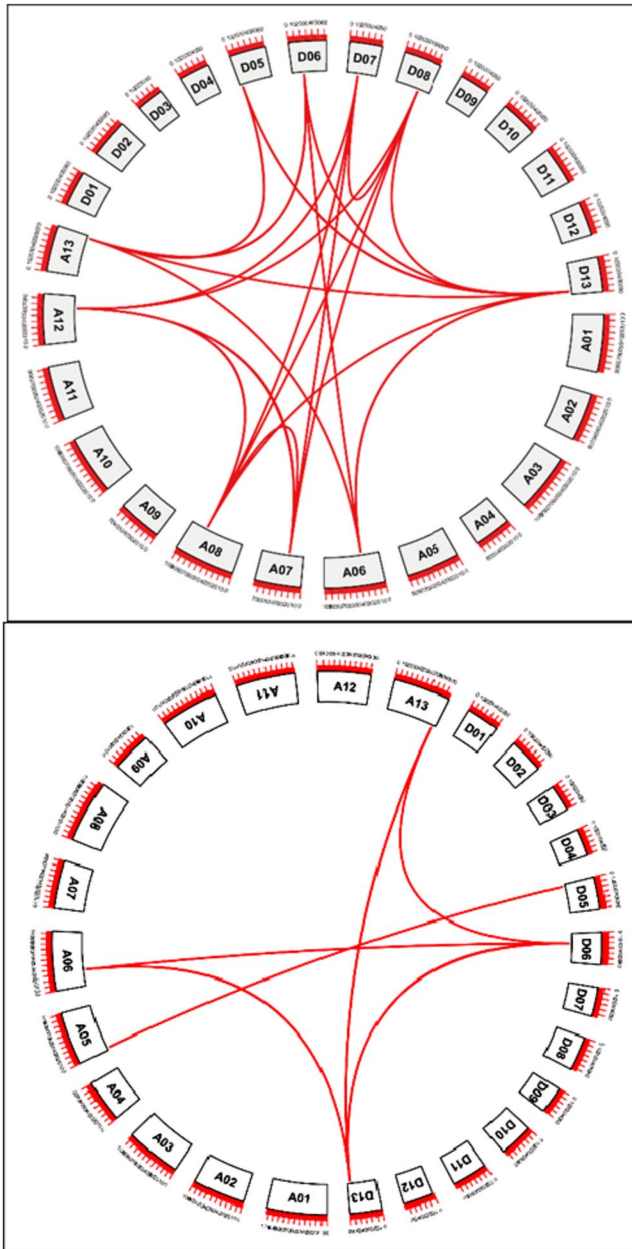


Fig. 3 Duplicated RCI2 gene pairs identified in GhRCI2 and GbRCI2 gene family

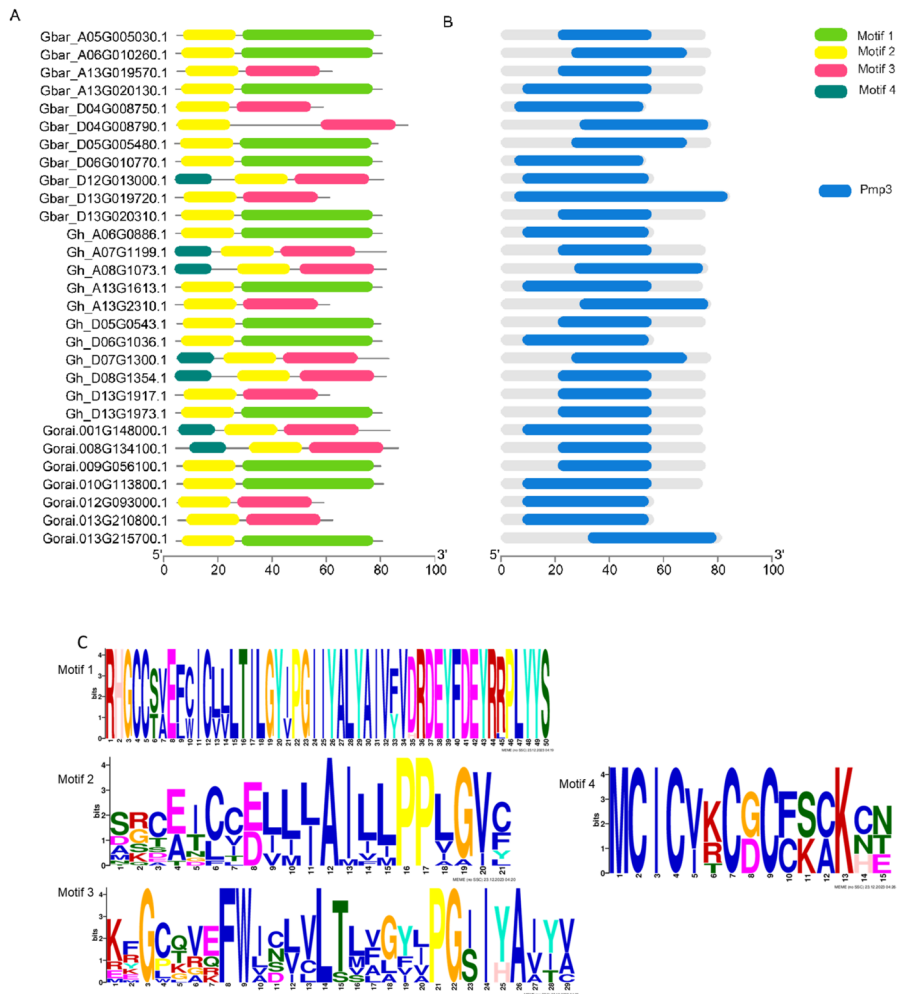


Fig. 4 Conserved motif and domain analysis of RC12 proteins across three cotton species: *Gossypium hirsutum*, *barbadense*, and *remondae*. **A** The distribution of conserved motifs among RC12 proteins was investigated using MEME and TB tools, revealing the identification of 10 conserved motifs in the cotton species. **B** The domain analysis of RC12 proteins is presented, providing insights into the structural characteristics of these proteins in the context of their conserved domains. **C** Distribution of amino acid positions of conserved motifs for RC12 proteins

Promoter Region Analysis of Cotton RC12 Genes

The study of *cis*-regulatory elements in the promoter regions of genes provides information on stress-responsive or tissue-specific gene expression patterns (Bilas et al. 2016). *cis*-regulatory factors in the promoter region and plays a key role in genes expression regulation. To study the possible expression profile of Cotton RC12 genes, we used online Plant CARE software to evaluate the 2.0-kb genomic regions

Subcellular localization

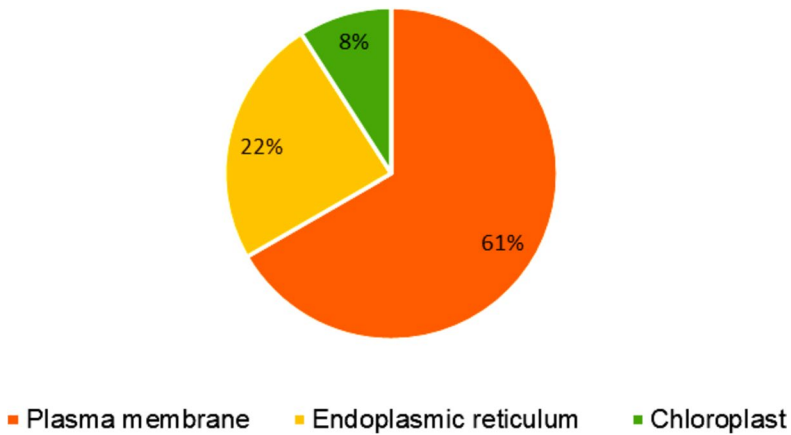


Fig. 5 Subcellular localization prediction analysis of RCI2 genes

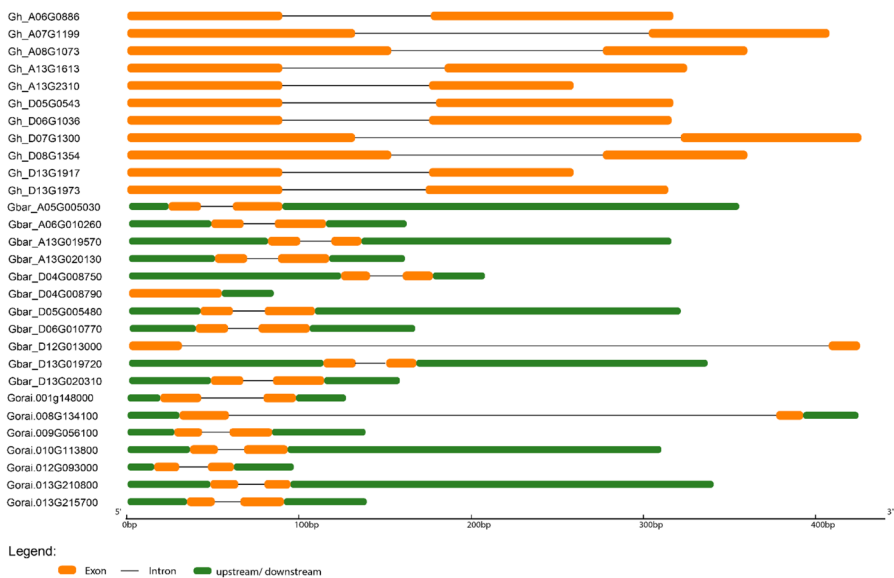


Fig. 6 Gene structures of the RCI2 genes in cotton, determined through alignment of CDS sequences with the corresponding genomic DNA (gDNA) sequences using the GSDS Server. In the representation, orange bars denote coding sequences (CDSs), green bars represent untranslated regions (UTRs), and plain lines illustrate introns, providing a comprehensive visualization of the gene organization within the cotton RCI2 genes

upstream of their starting codons. In the genomic sequences of cotton RCI2 genes, a variety of *cis*-regulatory factors associated with stress, hormones, and development were discovered. Hormone-related *cis*-elements, such as CGTCA-motif, ABRE, TGA-element, AuxRR-core, GARE-motif, ERE, and TCA-element were the most frequent, demonstrating their significance in response to various hormones such as ABA, SA, GA, auxin, and MeJA (Fig. 7). ABA hormones and Meja in their promoter regions expressed in all genes of *G. hirsutum* except few promoter sequences. All identified cotton RCI2 genes were enriched with light responsiveness elements. Most of the *cis*-regulatory elements were stress-responsive elements, including TC-rich repeats, W-box MBS, HSE, MBS, ARE, and WUN-motif, which are involved in wound responsiveness, defense responsiveness, heat stress responsiveness, drought inducibility, abiotic stress and anaerobic induction, respectively. However, from all of these genes included, three different types of *cis*-elements related to hormones were found in their promoter regions. Furthermore, the Cotton RCI2 promoter regions comprised 4 kinds of *cis*-elements related to development e.g., CAT-box, CCGTCC-box, O2-site, and Skin-1 motif (Fig. 7), which are associated with endosperm expression, meristem expression, meristem specific activation, and zein metabolism, respectively. These findings indicate that the Cotton RCI2 genes may be involved in tissue formation along with responses to various stresses and hormones.

Tissue-Specific Expression of RCI2 Genes in Cotton

By using online transcriptomic data, the expression levels of ChRCI2 genes in the several tissues and abiotic stresses such as heat, cold, salt, and drought was investigated. Interestingly, the expression of 10 ChRCI2 was undetectable in all the investigated tissues, suggesting that it would have some expression patterns. Gh_A13G2310 gene highly expressed in 10 dpa fiber, moderately expressed in petals

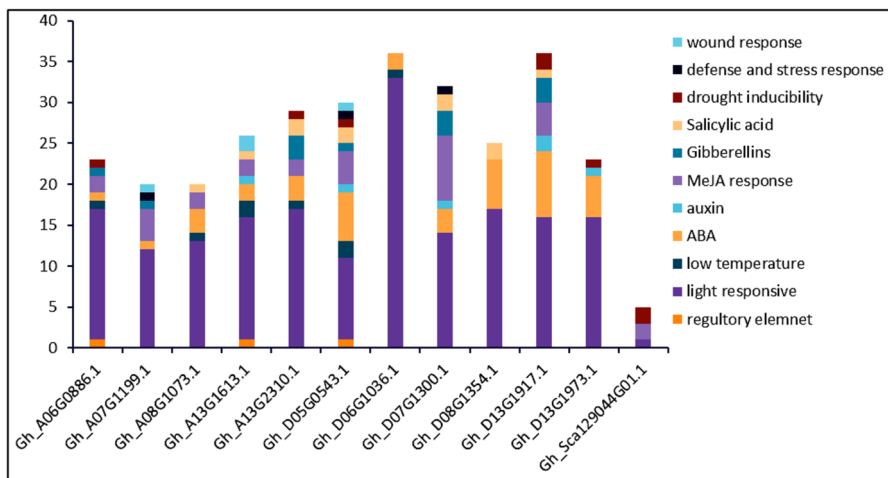


Fig. 7 Promoter analysis of GhRCI2 genes of cotton. The numbers of *cis*-elements (1 kb upstream of the translation start site) are presented in bar graphs while similar *cis*-elements are shown with similar colors

of the leaves and 50 dpa fiber, and also weakly expressed in stamen and caly- cle. Gh_D13G1917 showed higher expression pattern in petal and 10 dpa fiber, showed moderately expression in calycle tissues and weakly expressed in 50 dpa and stamen. Gh_D13G1917 was highly expressed at 3 h salt and cold treatment. It was also expressed at 12 h salt treatment and moderately expressed at 1 h heat and 6 h cold and also showed weakly expression at different intervals for different treatment. Gh_A13G2310 showed moderate expression at 3 h cold and 1 h heat and also lower transcription at 3 h salt and 12 h salt treatment (Fig. 8).

Expression Analysis of GhRCI2 After Cold, Salinity and ABA Stress

The expression patterns of four GhRCI2 genes were analyzed under cold, salt, and ABA treatments using qRT-PCR. As shown in Fig. 9, Gh_D13G1917, Gh_A13G2310 and Gh_D07G1300 displayed the highest expression after 3 h of salt and ABA exposure, aligning with transcriptomic data, before declining at 6 h and increasing again at 12 h (except for Gh_D13G1917 under salt stress). Under cold stress, Gh_D13G1917 and Gh_D08G1354 showed similar expression trends to salt/ABA at 3 and 6 h, while Gh_A13G2310 and Gh_D07G1300 exhibited opposite patterns. By 12 h of cold stress, expression of all four GhRCI2 genes further decreased. Overall, the qRT-PCR expression profiles of the cotton RCI2 genes were relatively comparable between salt and ABA treatments, whereas more variability was witnessed under cold stress across the analysis time points (Fig. 9).

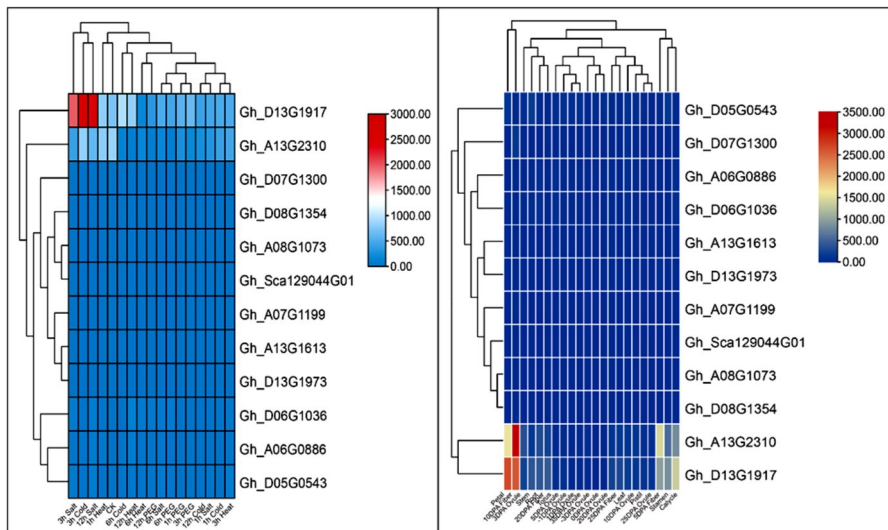


Fig. 8 Tissue Expression patterns of RCI2 genes of *Gossypium hirsutum* species. The RNA-seq data associated to tissue-specific expression were retrieved from CottonFGD database (<https://cottonfgd.org/>) and Genesis software was used to generate heatmaps

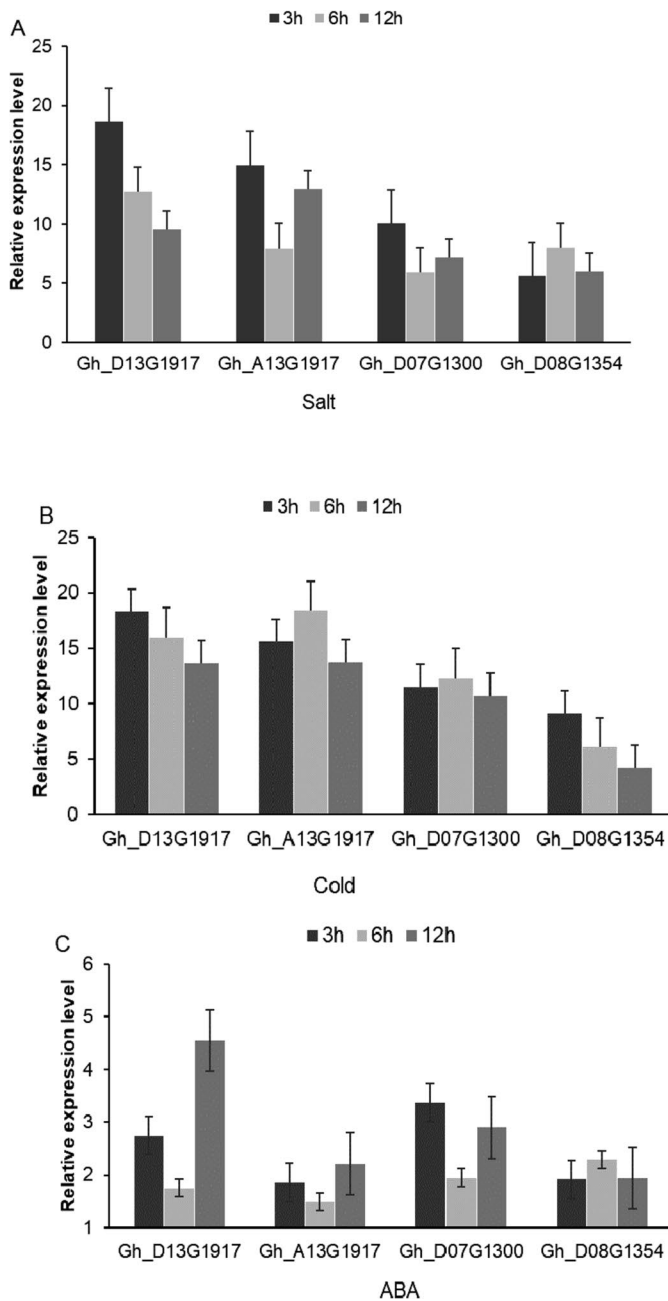


Fig. 9 Expression profiles of GhRCI2 genes in response to different stress conditions, namely salt (A), cold (B), and ABA (C). The relative expression levels of GhRCI2 genes were assessed at three time points: 3, 6, and 12 h of treatment, providing insights into the dynamic nature of gene expression under various stress stimuli

Discussion

RCI2 genes have been identified and characterized in a wide range of plant species including *Arabidopsis*, rice (Medina et al. 2007), maize (Zhao et al. 2014a, b), and *Medicago sativa* and *Medicago truncatula* (Long et al. 2015) but not yet in cotton. Cotton has high transpiration rate and vulnerable to several abiotic stresses (Liu et al. 2014). In this investigation, 36 RCI2 genes were discovered in the cotton genome and no tandem or segmental duplication events were identified among the cotton RCI2 genes (Table 1, Fig. 1), showing that the RCI2 gene family evolved slowly in cotton. Increasing evidence indicates that RCI2 genes play critical role in various abiotic responses particularly in regulating plants tolerance to cold, drought and salt stress (Medina et al. 2007). RCI2s ought to be a main focus in molecular breeding because abiotic stressors such as salt, drought, and cold have a significant impact on plant growth (Shinozaki et al. 2003). RCI2 genes family mostly represent a similar organization of intron-exon, which showed that RCI2 genes have been structurally conserved during evolution and encode small, hydrophobic proteins having 2 putative transmembrane domains. Most of the AtRCI2 genes localized in the plasma membrane of the plant cell (Zhang et al. 2022). Due to sequences similarity RCI2 genes seem to be evolved from a common ancestor. We were capable to conduct an organized examination of the RCI2 family in cotton because of the whole genome sequencing of this major crop species.

RCI2 associated genes present in cotton species have also been found in a wide range of plant species including maize, wheat, and barely (Zhao et al. 2014a, b; Imai et al. 2005; Goddard et al. 1993) and also in the *Physcomitrella patens* (Kroemer et al. 2004). RCI2 genes can function in membranes other than the plasma membrane (PM). Nearly RCI2 proteins, for example, have been discovered in the intracellular membranes, golgi apparatus, endoplasmic reticulum, punctate structures and in chloroplast thylakoid (Medina et al. 2007; Long et al. 2015).

When a plant is exposed to biotic and abiotic stresses, the plasma membrane acts as the first obstacle to be crossed, and evidence suggests that abiotic factors activate RCI2 gene expression in response to drought, cold, salt and ABA stress (da Silva et al. 2023; Kim et al. 2016; Chidambaram et al. 2015). For example, the expression pattern of AcPMP3-1 and 2 genes was significantly increased in response to a variety of abiotic factors e.g., drought, cold and salt (Rocha 2016; Inada et al. 2005). Most of the maize RCI2 genes exhibit distinct expression patterns in response to salt stress and drought stress (Zhao et al. 2014a, b; Fu et al. 2012). In this experiment, all Cotton RCI2 genes showed response to abiotic stresses (Fig. 8), and *cis*-elements related to stress were present in the promoter regions (Fig. 7), signifying their effects to various abiotic stresses. Associated results were found in *Pennisetum glaucum* (Yeshvekar et al. 2017), *Salvia miltiorrhiza* (Wang et al. 2016), and *P. tenuiflora* (Zhang et al. 2008).

To well understand the evolutionary relationships of RCI2 genes in different cotton species, we developed a Neighbor joining tree by using MEGA software. The phylogenetic analyses can provide a more comprehensive view of their

evolution–function linkages. Clade A and B represent most of the genes from cotton species and represent salt and low temperature responsive gene family (Fig 1). While clade 3rd represents low temperature membrane. Most of the RCI2 genes in cotton species represent low temperature and salt responsive genes while few genes represent hydrophobic proteins LTI6A and LTI6B genes, early initiation by abscisic acid and low temperature while late stimulation of this hydrophobic protein by drought and salt stresses. Furthermore, only 6 genes represent plasma membrane proteins 3 which include 3 genes from *G. hirsutum*. Since phylogenetic tree, RCI2 genes from *Caenorhabditis elegans* (T23F2.3, T23F2.5) *Saccharomyces cerevisiae* (PMP3) is distantly related to those from plant species, regardless of the fact that the transmembrane regions of most of these proteins are highly conserved. The conserved motif analysis also helped to support the evolutionary tree. The phylogenetic study of the cotton RCI2 genes provides a valuable benchmark for future cloning and valuable functional study.

Cotton RCI2 gene structural analysis revealed that they showed similar intron/exons structure, which is same with previous research in Arabidopsis, rice, cucumber and maize. The process of structural diversity of genes and evolution of conserved motif allows for the formation of multigene families (Wang et al. 2013a, b). RCI2 genes have been extremely conserved structurally throughout evolution, with the majority of them consisting of two exons and single intron (Medina et al. 2007). In this study, we found 2 exons and one intron in all cotton RCI2 genes similar with Arabidopsis, rice, maize and cucumber introns/exons structures. The number of introns in genes responding to abiotic and biotic conditions is also found to be lower, as previously shown in the poplar and cotton LEA gene family (Magwanga et al. 2018; Lan et al. 2013).

A unique pattern was discovered by mapping 36 cotton RCI2s on cotton chromosomes. They were mostly found at the end of many chromosomes but no presence on chromosomes 4, 5, or 6. Despite the fact that the number of cottons RCI2s was more than that of Arabidopsis and the gene duplication patterns in the Arabidopsis RCI2 family were distinctly different (Medina et al. 2007). Among 8 AtRCI2s in Arabidopsis, two tandem duplications events (AtRCI2A, AtRCI2B and AtRCI2E, AtRCI2F), and one segmental duplication (AtRCI2D, AtRCI2E) event were found (Medina et al. 2007). It is important to note that the rice genome has a record of 12 RCI2 (OsRCI2-1–12) genes, although only OsRCI2-3 and OsRCI2-4 found as tandem duplication event (Medina et al. 2007). Though, RCI2 genes in cotton, Arabidopsis and rice are different, the expansion method of RCI2 gene may differ in the three species through evolution. Moreover, sequencing analysis revealed that all these RCI2 genes appear highly conserved across species, indicating that the RCI2 genes in plant genomes is an evolutionarily conserved domain. Gene duplication is a characteristic feature of genome architecture, which insights an important role in plant genomic evolution, resulting the formation of new raw genetic materials that can be influenced by selection, mutation and genetic drift, ultimately evolves with new gene functions and gene interacting networks (Flagel and Wendel 2009). Adaptability and evolution of plants is ensured by gene duplication which leads to the genome expansion

with diversified gene functions (Xu et al. 2012). The creation of new gene functions resulted due to gene duplication event during the evolution of cotton (Rong et al. 2010).

Cotton RCI2 genes promoter analysis showed many *cis*-elements associated with plant growth, abiotic stresses and phytohormones responses. Various studies have found that light has a significant influence on plant development, growth, and differentiation (Fahad et al. 2023a, b). Several important elements such as ABA responsiveness elements ABRE (Narusaka et al. 2003), stress-responsive factors T-rich repeats (Diaz-Martin et al. 2005), GCC-boxes (Song et al. 2005), gibberellins responsive elements TATC-box, and wound responsive elements Wun-motif were identified. N-terminal repeat motifs also reacted to low temperatures response (Maestrini et al. 2009), however TCA and CGTCA motifs were shown to be involved in response to methyl jasmonate and SA elements, respectively (Wen et al. 2014). W-boxes and ABRE were also found in the promoter areas of several genes, which mediated responsiveness to drought stress and ABA (Singh et al. 2002). The discovery of various light-responsive motifs in the cotton RCI2 promoter regions indicated that light signaling modulates RCI2 gene expression. The presence of several *cis*-elements associated with abiotic stress and phytohormones responses further revealed the functions of cotton RCI2 genes family.

Conclusion

Genome-wide analysis was performed in this study to find 36 Cotton RCI2 genes from cotton genome. The cotton RCI2 family genes were described into 6 subgroups based on their phylogenetic tree analysis. Gene structure and protein motif distribution are consistent in proteins of the same subgroups. According to chromosomal mapping, cotton RCI2 genes were found to be dispersed irregularly over the whole cotton genome. The evolutionary relationships, motif distribution, structural properties, duplication events, chromosomal mapping, and expression pattern of RCI2 genes in cotton were all thoroughly investigated. Furthermore, all of the cotton RCI2 genes were highly sensitive to drought, ABA, salt, and cold treatments, demonstrating that they may be employed as genetic objects to produce stress-resistant plants. The results of this study provide a solid base for further functional identification and characterization of RCI2 family in cotton.

Author Contributions MF, MTA, MS, MA, WL, IU, IU, HIM and AB conceived and designed the project. MF, MTA, MS, MA, WL, IU, IU, HIM and AB performed the experiments. MF, MTA, MS, MA, WL, IU, IU, HIM and AB wrote the manuscript. HIM, MF and LT revised and edited the manuscript. All authors read and approved the final version of the manuscript submitted for publication.

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Availability of Data and Materials Not applicable.

Declarations

Conflict of interest The authors declare that they have no conflict of interests.

Ethical Approval This article does not contain any studies with human participants or animals performed by any of the authors.

Consent to Participate Not applicable.

Consent for Publication Not applicable.

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Authors and Affiliations

Muhammad Fahad¹ · Leeza Tariq² · Muhammad Tanveer Altaf³ · Muhammad Shahnawaz⁴ · Mudassir Aslam⁴ · Waqas Liaqat⁵ · Inayat Ullah⁶ · Izhar Ullah⁷ · Heba I. Mohamed⁸  · Abdul Basit⁹

✉ Heba I. Mohamed
hebaibrahim79@gmail.com

- ¹ State Key Laboratory of Rice Biology, College of Agriculture and Biotechnology, Zhejiang University, Hangzhou 310058, Zhejiang, China
- ² National Key Laboratory for Rice Biology, Institute of Biotechnology, Zhejiang University, Hangzhou 310058, China
- ³ Department of Plant Production and Technologies, Faculty of Agricultural Sciences and Technologies, Sivas University of Science and Technology, 58140 Sivas, Turkey
- ⁴ Department of Plant Breeding and Genetics, University College of Agriculture, Bahauddin Zakariya University, Multan, Pakistan
- ⁵ Department of Field Crops, Faculty of Agriculture, Institute of Natural and Applied Sciences, Çukurova University, 01330 Adana, Turkey
- ⁶ Department of Agricultural Mechanization and Renewable Energy Technologies, The University of Agriculture Peshawar, Peshawar 25130, Pakistan
- ⁷ Department of Horticulture, Faculty of Agriculture, Ondokuz Mayıs University, Samsun, Turkey
- ⁸ Department of Biological and Geological Sciences, Faculty of Education, Ain Shams University, Cairo 11341, Egypt
- ⁹ Department of Horticulture, Kyungpook National University, Daegu 41566, South Korea